Pumping Lemma for RE :

$W = \ldots$

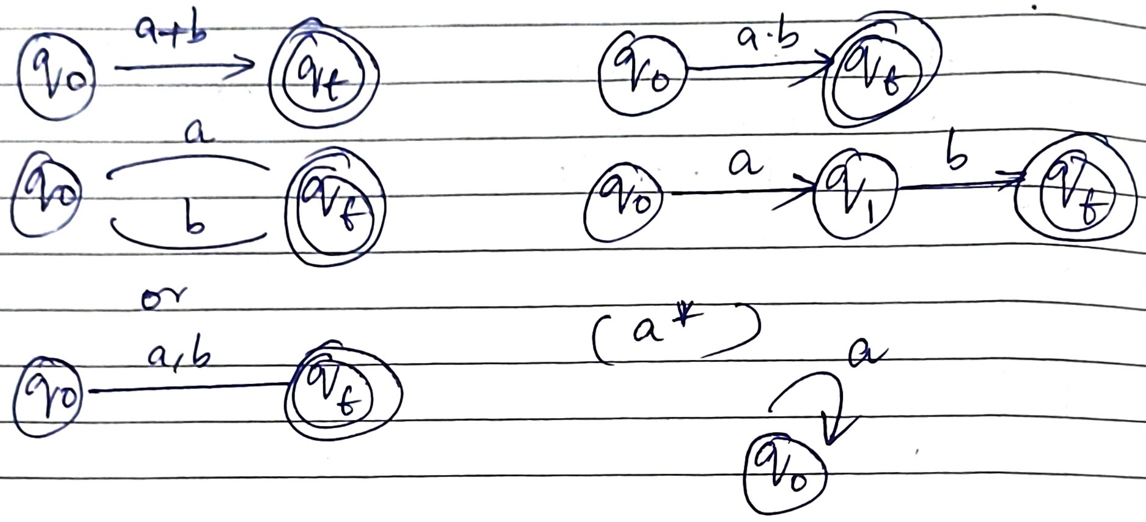
$W = xyz$

for $\forall$ $\boxed{k \geq 0}$, $W = xy^kz \in L$

$\therefore$ if condition = satisfied

then ; it is RE .

RE $\rightarrow$ DFA : eg: $(a+b)$: $(a \cdot b)$



DFA - Minimization :

* Remove non-reachable states.

* Check Equivalence classes.

→ 0 Equivalence

→ 1 Equivalence

→ 2 Equivalence

. . . .

* Combine & construct the DFA for the final equivalent states.

Ambiguity In Grammar : * If n(parse trees) > 1, the grammar G is Ambiguous.

Q. Consider the G with production rule, ...

$E \rightarrow I$

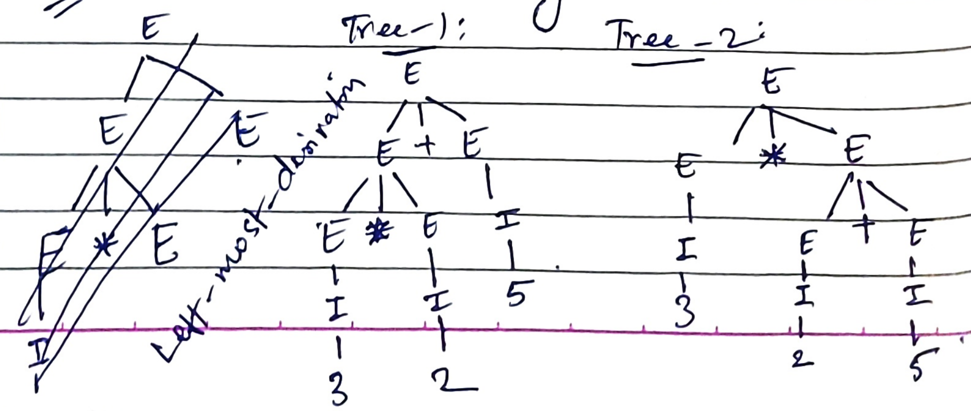
$E \rightarrow E+E$

$E \rightarrow E*E$

$E \rightarrow (E)$

$I \rightarrow \varepsilon | 0 | 1 | \ldots$

Sol Lets derive string "3*2+5" !



CNF Chomsky Normal Form	GNF Griebach Normal Form
$S \rightarrow \varepsilon$	$S \rightarrow \varepsilon$
$A \rightarrow a$	$A \rightarrow a$
$A \rightarrow AB$	$A \rightarrow a\,ABC\ldots$

A = (caps) = Non-terminal
a = Terminal.
S = Start state.

= CFG (context free Grammar).

⇒

Module - 1

Finite - automaton (machine) → Transition diagram, Graph, Function.

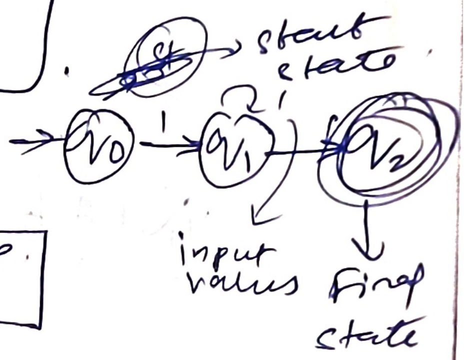
$\{Q, \Sigma, q_0, F, \delta\}$

Intermediate result to prove a larger result = "Lemma".

[Design & analysis of complex Hardware & software system.]

- Finite Automata ⇄ 5 tuples
  - without O/P
    - DFA
    - NDFA
    - ENFA
  - with O/P
    - Mealy
    - Moore

    machines



DFA → single path determinism

no. of state = |min. string| + 1

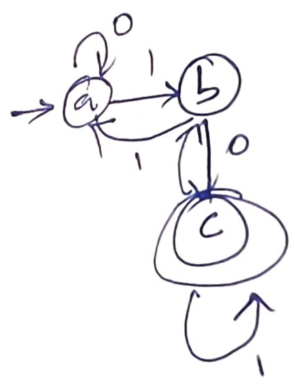
Examples :

DFA = $Q = \{a, b, c\}$ $q_0 = \{a\}$

$\Sigma = \{0, 1\}$ $F = \{c\}$

Table ;

Present state	0	1
→ a	a	b.
b	c	a
* c	b	c



DFA	NFA
$\delta : Q \times \Sigma \rightarrow Q.$	$\delta : Q \times \Sigma \rightarrow 2^{Q}$

→ Module-1 :

Automaton = Abstract computing device / A machine.

FA = Finite Automata.

- With O/P
  - DFA
  - N-FA
  - ε-NFA
- Without O/P
  - Mealy
  - Moore machines.

5 tuples:

$= (Q, \Sigma, q_0, F, \delta)$

- Q → states
- Σ → Alphabets
- q_0 → Initial state
- F → Final state
- δ → Transition function.

DFA ~~&NFA~~ $\boxed{\delta : Q \times \Sigma \rightarrow Q}$

NFA with ε $\boxed{\delta : Q \times \Sigma \cup \varepsilon \rightarrow \cancel{Q}\ 2^{Q}}$

NFA $\boxed{\delta : Q \times \Sigma \rightarrow 2^{Q}}$

⇒ pumping lemma for regular languages :

$W = \ldots$

$W = xyz$

For $k \geq 0$, $xy^{k}z \in L$.

(1) Design DFA : $\Sigma = \{0, 1\}$

accepts even no. of 0's & even no. of 1's.

should be compulsory

$$\text{total states} = |\text{min length}| + 1$$
$$= 2 + 1 = \boxed{3}$$

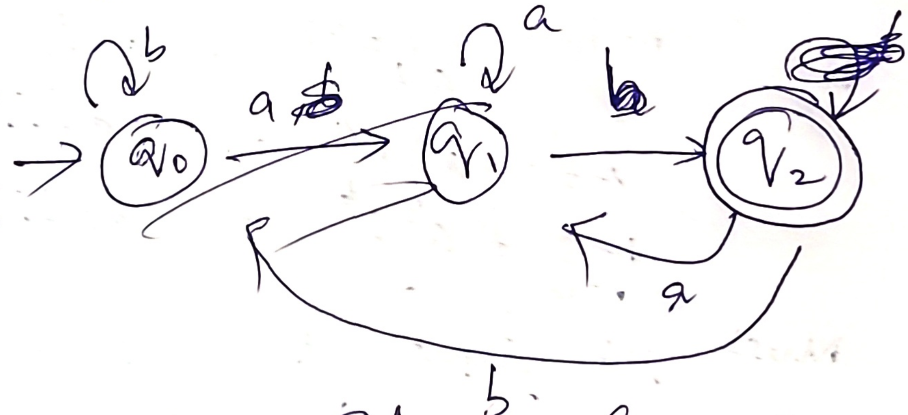
(2). Construct DFA, accepting all strings over $\{a, b\}$ ending with 'ab'?

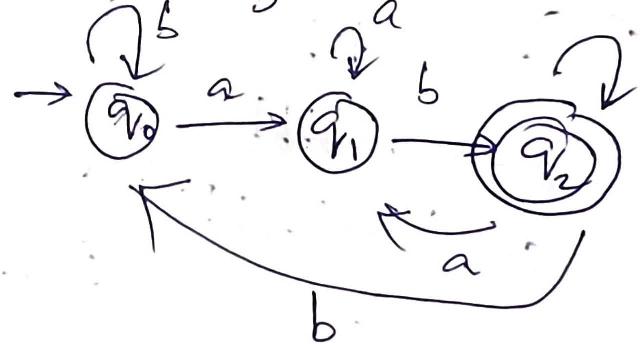
~~$L = \{a, b\}$~~ $\Sigma = \{a, b\}$

$L = \{ab, aab, abab, bbab, \ldots\}$

$$\text{states} = |\text{min length}| + 1$$
$$= 2 + 1 = 3$$

ab
bab





Given : NFA → DFA : $\delta : Q \times \Sigma \rightarrow Q$

$\delta : Q \times \Sigma \rightarrow 2^Q$ $\cancel{\delta \times \Sigma} \rightarrow$

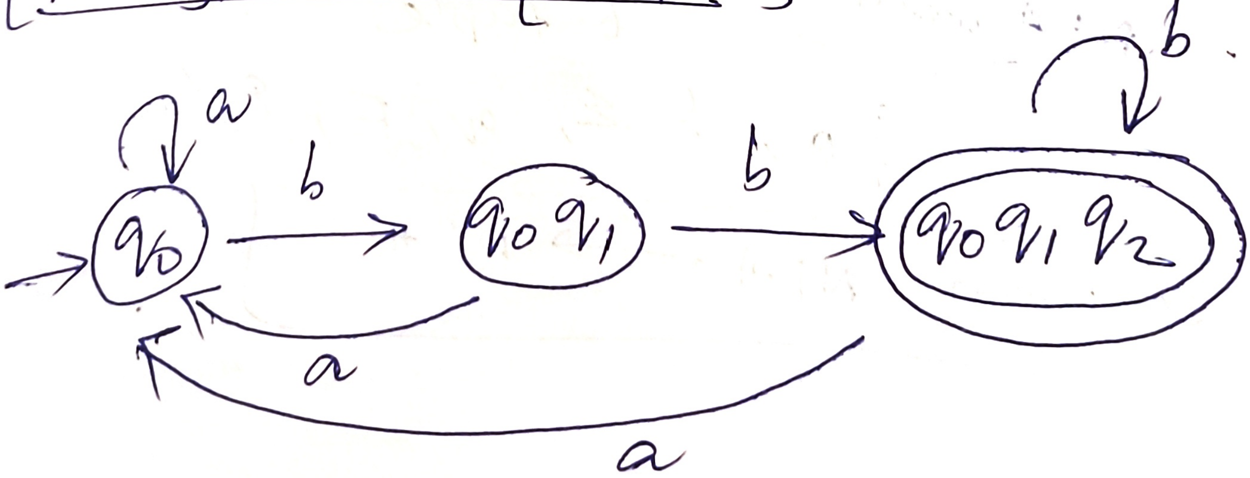
1st step : Construct NFA transition table.

⇒ NFA

δ	a	b
→ q_0	q_0	$\{q_0 q_1\}$
q_1	ε	q_2
* q_2	ε	a

⇒ Transition table NFA.

δ	a	b
→ q_0	q_0	$[q_0 q_1]$
$[q_0 q_1]$	q_0	$[q_0 q_1 q_2]$
$[q_0 q_1 q_2]$	q_0	$[q_0 q_1 q_2]$



~~CFG~~ Module-2

* Context Free Grammar [CFG] — 4 Tuples

- To generate all possible ~~strings~~ patterns of strings in given formal language.

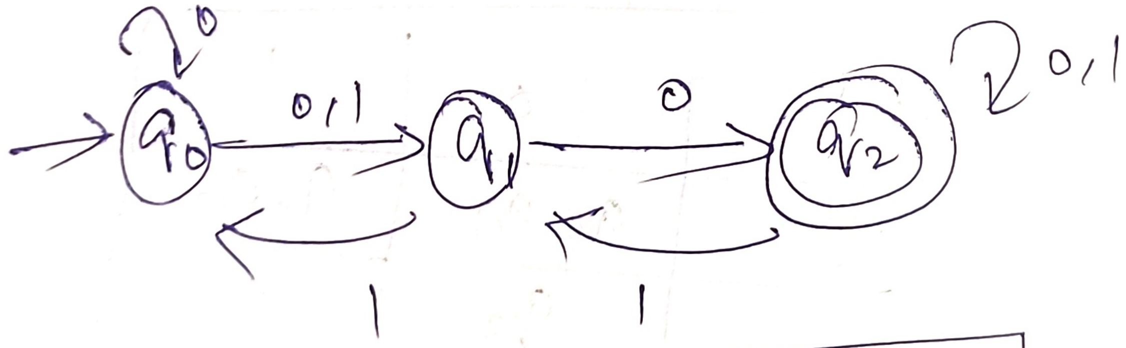
$G = \{N, T, P, S\}$

- N → Non terminal → UPPER-CASE
- T → Terminal → LOWER CASE
- P → Production rule
- S → Start symbol

NFA :-

$Q = \{q_0, q_1, q_2\}$

$\Sigma = \{0,1\}$ $\quad q_0 = \{q_0\}, F = \{q_2\}$



δ	0	1
→ q_0	$q_0\ q_1$	q_1
q_1	q_2	q_0
* q_2	q_2	$q_2\ q_1$

(P) . Construct NFA where

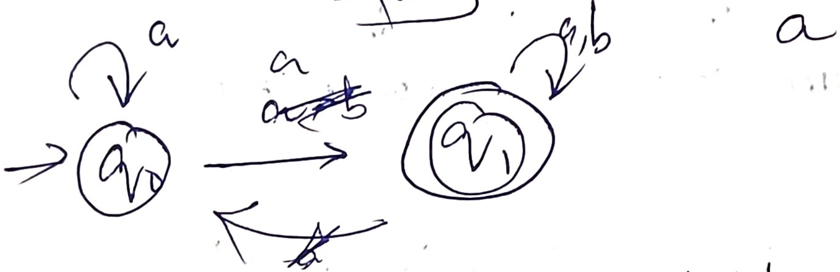
$l = \{$ start with 'a' $\}$

$\Sigma = \{a, b\}$

$L = \{a, ab, abb, aba, abbb, abca, \}$

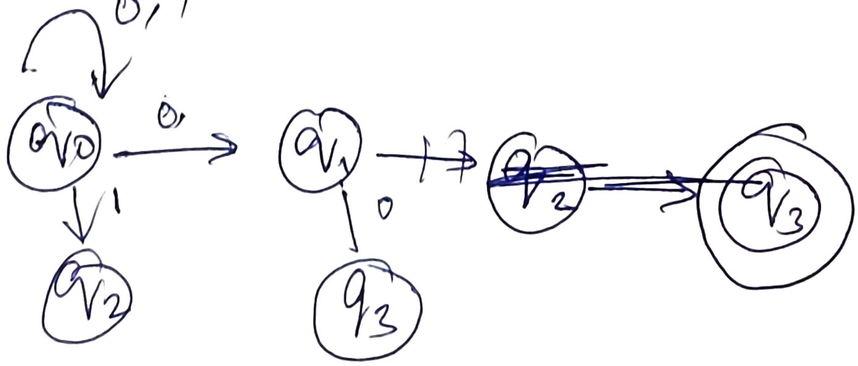
no. of states = |min length| + 1

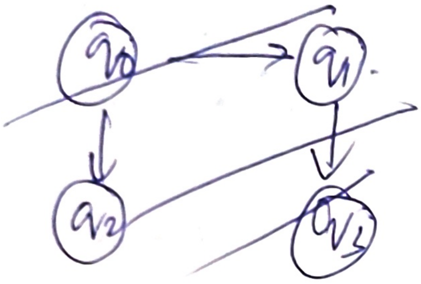
$= \boxed{2}$

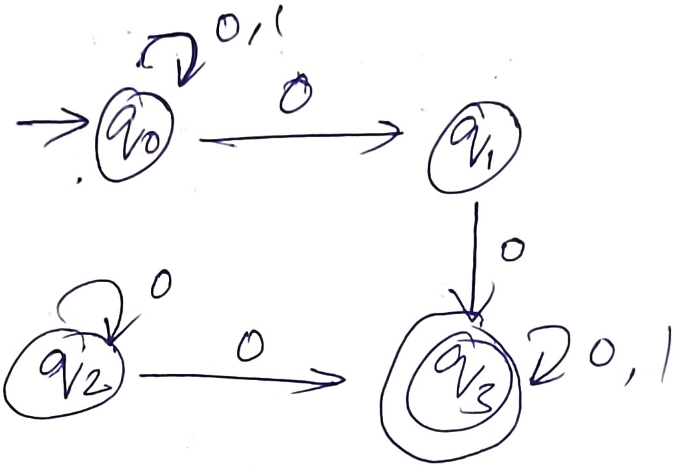


(P) Design NFA for transition table

State	0	1
→ q_0	$q_0\ q_1$	$q_0\ q_2$
q_1	q_3	ε
q_2	$q_2\ q_3$	q_3
* q_3	q_3	q_3







State	0	1
→ q_0	q_0, q_1	q_0, q_2
q_1	q_3	ε
q_2	q_2, q_3	q_3
* q_3	q_3	

Construct NFA that accepts all strings over $\{0,1\}$, of length 2.

$\Sigma = \{0,1\}$.

$L = \{00, 01, 10, 11\}$.

Sol no. of state = |min length| + 1

= 3.

NFA with ε: 6 tuple

$\{Q, \delta, \Sigma, q_0, F, \varepsilon\}$.

$\boxed{\text{Conversion NFA} \longrightarrow \text{DFA}}$

Regular Expressions :-

(U) Union, (*) Kleen, (+) Plus, (?) optional

Q) Write RE for language accepting all strings containing any no. of a's and b's:

$\boxed{RE = (a+b)^*}$

Starting with 1 and ending with 0 over $\Sigma = \{0, 1\}$.

First symbol is 1, Last symbol is 0.

$(1+0)^*$

Q) Start with a but not having consecutive b's.

$L = \{a, aba, aab, aaa, \ldots\}$

$RE = \{a + ab^*\}$

CFG : Context Free Grammar.

$G = \{V, T, P, S\}$ or $\{N\,T\,P\,S\}$

V → Variables (non-terminals)

T → (terminals)

P → Production rule

S → Start symbol

Q) Construct CFG for the Language no of a's over set $\Sigma = \{a\}$.

~~$\Sigma = \{a\}$~~ $L = \{\varepsilon, a, aa, aaa, aaaa, \ldots\}$

$RE = \{a^*\}$.

Production rule :

$S \to aS$ — (1)

$S \to \varepsilon$ — (2)

Derive input "aaa".

S

aS → $S \to aS$

$aa(aS)$ → $S \to aS$

~~$aaa(aS)$ → $S \to aS$~~

$aaaa$ aaa $S \to \varepsilon$

① Construct CFG.

$L = \{ wcw^R \mid \text{where } w \in (a+b)^* \}$.

string — string (reversed).

$G = \{V, T, P, S\}$

$L = \{a, b, ab, aabb, \varepsilon\}, \ldots$

$L = \{aacaa, bbcbb, abcba, \cancel{aab}, \ldots\}$

Production rule:

$S \rightarrow asa$ — ①
$S \rightarrow bsb$ — ②
$S \rightarrow c$ — ③

derive input: abcba.

S
a S a → $S \rightarrow bsb$.
a b S b a
$S \rightarrow c$
$\boxed{abcba}$
- input derived.

② $\cancel{S \rightarrow aSb} \mid \varepsilon$. CFG for $L = \{a^n b^n \mid n \geq 1\}$

$L = \{ab, aabb, aaabbb\}$

Production rule: $S \rightarrow ab$ — ①
$S \rightarrow \varepsilon$ — ②.

Derive input string:

input string: $S \rightarrow aaaabbbb$

S
a S b — $S \rightarrow aSb$.
a (a S b) b.
a a (a S b) b b
aaa(a S b) bbb
$S \rightarrow \varepsilon$
aaaa S bbbb = aaaabbbb

Input string is derived.

Derivation of CFG. *

* Sequence of . Production rules to get an i/p strings. =

2 - decisions.

Which non-terminal has to be replaced ~~first~~

Decide production rule based on the replaced non ~~terminal~~.

*

PDA : ("Finite state" + "Stack") machine

Input tape

A control unit

Stack (size = ∞)

Pumping lemma for Regular Languages. :-

(1) prove that $L = \{0^n 1^n \mid n \geq 1\}$ is not regular.

$w = 0^n 1^n$

Let $n = 1$ 01

$n = 2$ 0011.

Let

$w = xyz$.

$x = 0$ $z = 11$.

$y = 0$

Now, check

For $\forall k \geq 0$, $xy^kz \in L$.

$k = 2$ $xy^kz \in L$.

~~00~~

$xyyz \in L$.

$00011 \notin L$:=

∴ It is not ~~regu~~ RL.

* Ass L is regular
* Let n be a constant
* Let $w = xyz$, such that,
  1. $y \neq \varepsilon$.
  2. $|xy| \leq n$.
  3. For all $k \geq 0$, $xy^kz \in L$.

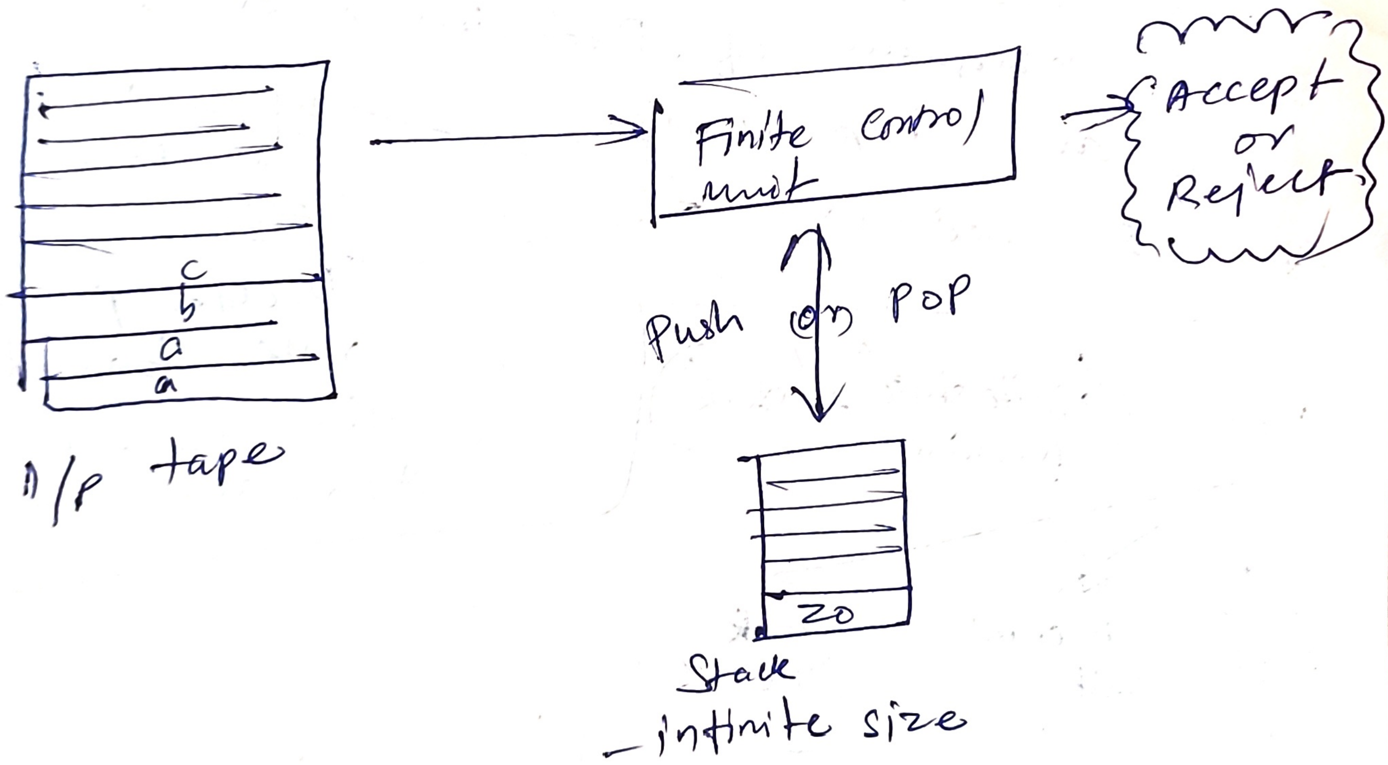
$w = 0^n \& 1^n$.

split $w = xyz$

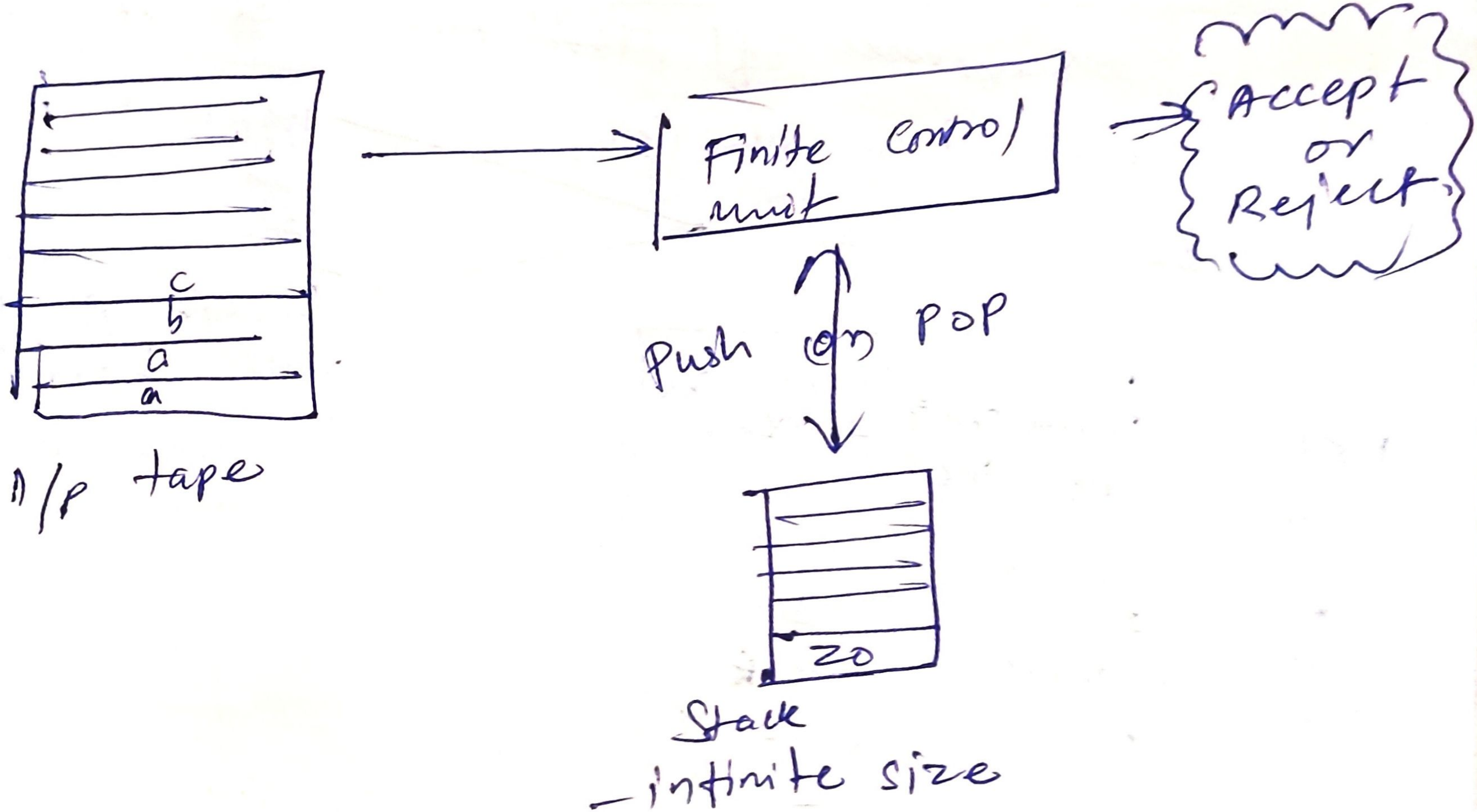
$y \neq \varepsilon$

$|xy| \leq n$.

For all $k \geq 0$, $xy^kz \in L$.



c
b
a
a
I/P tape
Finite control unit
Accept or Reject
Push or Pop
Z0
Stack
infinite size



Finite control unit
Accept or Reject
c
b
a
a
Push or pop
I/p tape
z0
Stack
- infinite size

① M- Divisible by 3. There is no final state.

Logic: 0 → remainders
1 → "
2 → " } 3 states

Transition table.

		0 3 6 9	1 4 7	2 5 8
0	q_0	q_0	q_1	
1	q_1			
2	q_2			

$\frac{0}{3} = 0$

$\frac{1}{3}$

3)10(3

3)4(1

3)5(1

		0	1	2	3	4	5	6	7	8	9	10	11
0	q_0	q_0	q_1	q_2	q_0	q_1	q_2	q_0	q_1	q_2	q_0	q_1	q_2
1	q_1	q_1	q_2	q_0	q_1	q_2	q_0	q_1	q_2	q_0	q_1	q_2	q_0
2	q_2	q_2	q_0	q_1	q_2	q_0	q_1	q_2	q_0	q_1	q_2	q_0	

$\frac{0}{3} = 0$

		0,3,6,9	1,4,7,10	2,5,8,11
0	q_0	q_0	q_1	q_2
1	q_1	q_1	q_2	q_0
2	q_2	q_2	q_0	q_1

3)20(6
18
2

$\frac{12}{3}$ 3)12(4
12

$\frac{1}{3}$ = 3)10

$\frac{1}{3} = \frac{2}{3}$

3)20(6
18
2

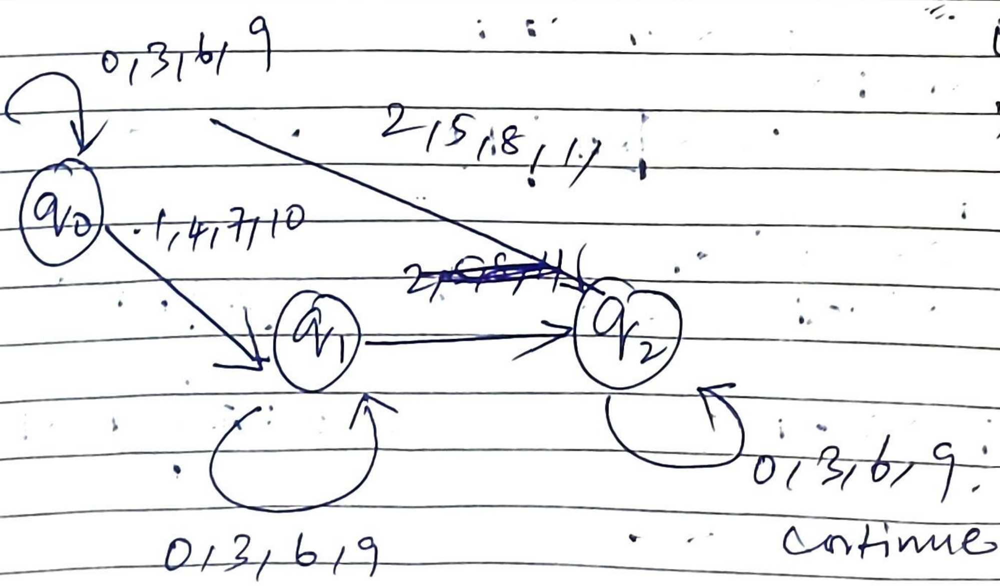
3)9(3
9

4)3

$\frac{3}{4}$ $\frac{2}{4}$

4)20

3)4(1
3
1



continue.

PAGE NO.
DATE

$a^n b^n$

aaabbb

$S \to AB$
$A \to aA \mid bB \mid b$
$B \to b$

~~$A \to a$~~ ~~$S \to A$~~
$S' \to A$

$S \to aA \mid bB \mid b$

$S \to aAbB$

$S \to AE$
$A \to AB$
$A \to a$
CNF

$S \to E$
$A \to A$
$A \to aABC \cdots$
GNF

$S \to aB \mid A \mid E \mid$

$aB \mid E \mid$

$S \to A$

$S \to aAB \mid bBB \mid bB$
$A \to aA \mid bB \mid b$
$B \to b$

$S \to CA \mid BB$
$B \to b \mid SB$
$C \to b$
$A \to a$

$S \to bA \mid BB$

$S \to ba \mid BB$
$S \to ba \mid b$

$\alpha = BB$

$\underline{\underline{\text{Ans}}}$: $S \to CA \mid BB$
$C \to b$;
$S \to bA \mid BB \Rightarrow S \to bA \mid bB \mid SBB$

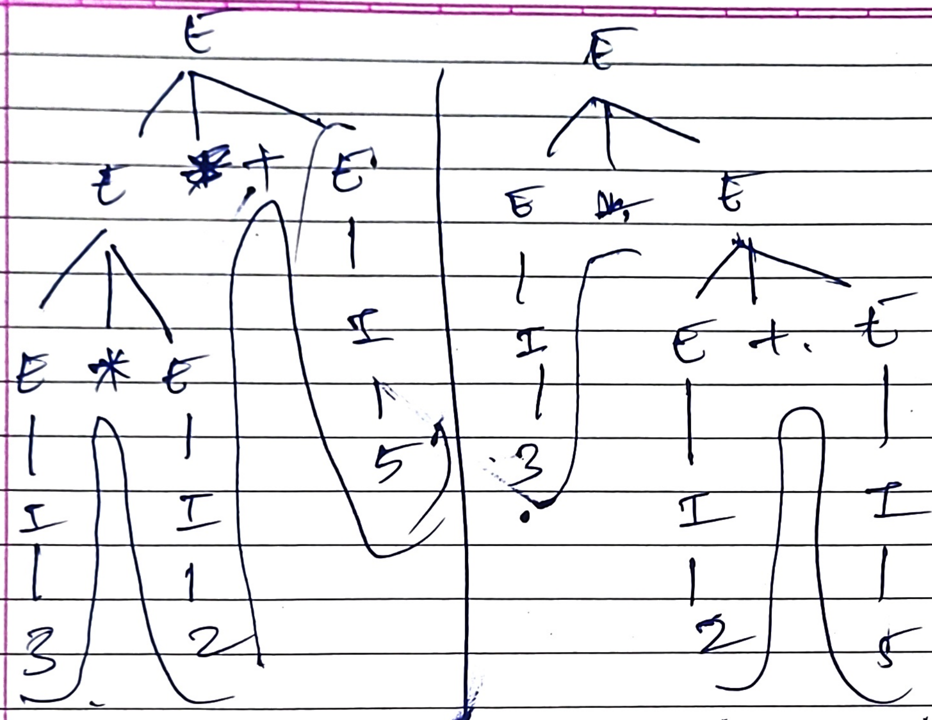
$S \to b \mid SB$
$B \to b \mid SB \longrightarrow S \to bA \mid bB$

$A \to A\alpha \mid \beta$
$\Downarrow$
$A \to \beta A'$
$A' \to \varepsilon \mid \alpha A'$

$B \to b \mid SB$;

$S \to bA \mid B \to b \mid bAB \mid bBB$

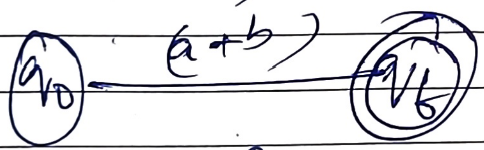
E $(3*2+5)$



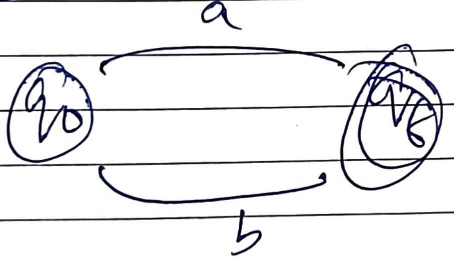
answer

RE: $(a+b)^*$

step-1

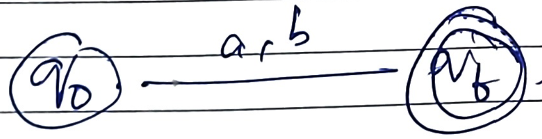


step-2

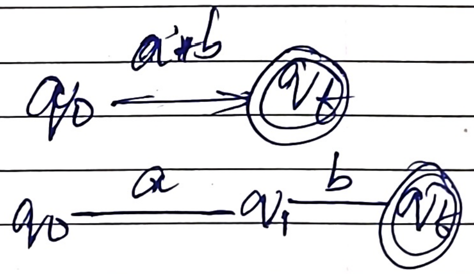


or

Step



$a*b$



a^* aa, aaa..